



RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ELECTRONICS ENGINEERING (VLSI D&T)

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SEMESTER IV

EV23421 – MICROPROCESSORS & MICROCONTROLLORS LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	SMART STREET LIGHT AUTOMATION SYSTEM
DATE OF SUBMISSION	04/05/2026
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Signature of HoD

Abstract :

Street lighting consumes a large amount of electrical energy, especially when lights remain ON during daytime due to manual control. This leads to energy wastage and increased electricity cost.

To overcome this issue, this project proposes a Smart Street Light Automation System using sensors and a microcontroller. The system automatically turns ON street lights during night and OFF during daytime using an LDR (Light Dependent Resistor).

Additionally, motion sensors can be used to increase brightness only when vehicles or people are detected, reducing power consumption further. This system is cost-effective, energy-efficient, and suitable for smart city applications.

Introduction :

Street lighting is very important for safety and visibility during night time. It is used in roads, streets, and public places to help people and vehicles move safely. In many places, street lights are still controlled manually or by using timers. This is not efficient because lights may remain ON during daytime, which leads to wastage of electricity.

Manual control also needs human effort and regular checking, which can cause mistakes. To solve this problem, a smart street light automation system is used. This system uses a sensor to detect light and a microcontroller to control the street lights automatically.

The system turns ON the lights during night and turns them OFF during daytime without human help. This helps in saving energy, reducing effort, and improving efficiency. It is a simple and useful solution for modern lighting systems.

Problem Statement :

In many areas, street lights are controlled manually or by using fixed timers. This method is not efficient because lights may remain ON during daytime, which leads to wastage of electricity and increased cost. Manual operation also requires human effort and regular monitoring, which can cause errors.

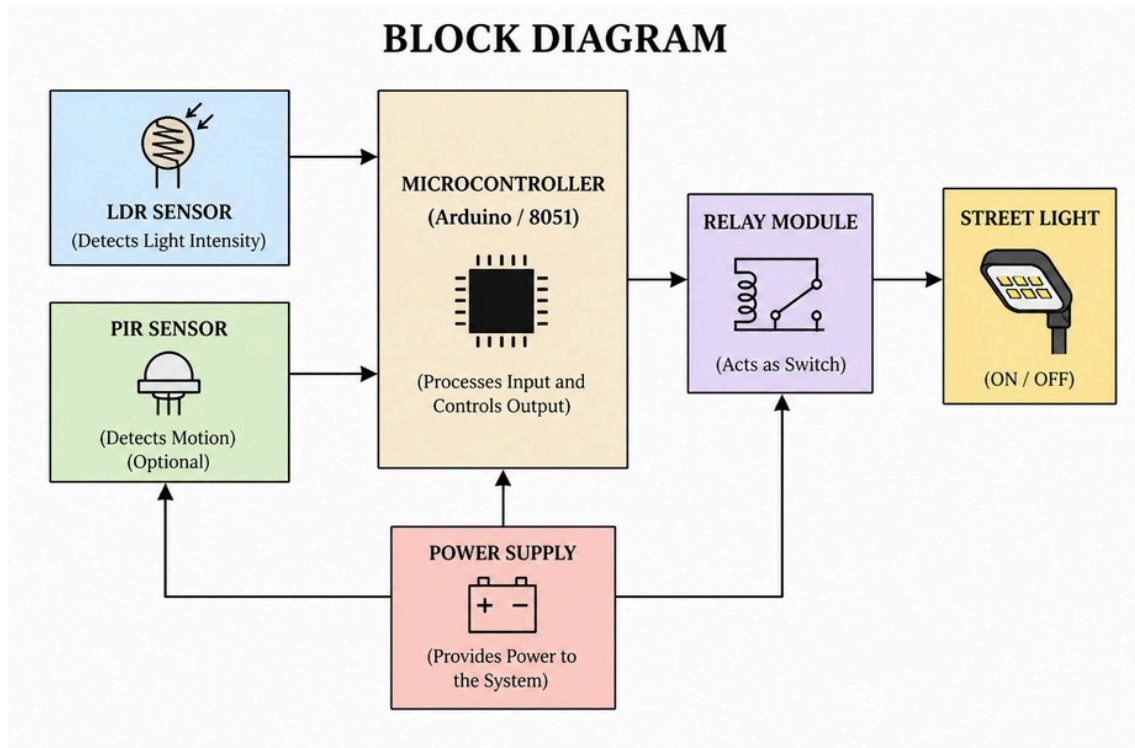
These systems cannot adjust based on changing light conditions like day and night. To solve this problem, an automatic system is needed that can detect light intensity and control the street lights automatically. The system should turn ON the lights during night and OFF during daytime without human involvement, ensuring energy saving and better efficiency.

Objectives :

The main objective of this project is to design and develop a smart street light automation system that works without human intervention. The system aims to automatically turn ON the street lights during night and turn them OFF during daytime by detecting the surrounding light intensity using a sensor. It also focuses on reducing energy consumption, minimizing manual effort, and improving the overall efficiency of the lighting system. Additionally, the project aims to provide a simple, cost- effective, and reliable solution that can be used in modern smart city applications.

Components Used :

LDR Sensor	– Used to detect light intensity (day/night)
Micr ocontroller (Arduino/8051)	– Controls the entire system
Relay Module	– Acts as a switch to turn lights ON/OFF
LED / Street Light	– Provides lighting output
Power Supply	– Supplies required Electrical Power
PIR Sensor (Optional)	– Detects motion of vehicles or people
Connecting Wires	– Used for circuit connections

Block Diagram:**Explanation :****1.LDR Sensor:**

The LDR (Light Dependent Resistor) is used to detect the intensity of light.
 It senses whether it is day or night.
 It sends the signal to the microcontroller based on light intensity.

2.PIR Sensor (optional):

Detects movement of vehicles or people
 Sends signal when motion is detected
 Helps in saving energy by controlling brightness

3. Microcontroller (Arduino/8051) :

Acts as the brain of the system
 Receives input from sensors
 Controls ON/OFF operation of street lights

4. Relay Module:

- Works as an electronic switch
- Receives signal from microcontroller
- Controls power supply to street light

5. Street light(LED):

- Output device of the system
- Turns ON during night
- Turns OFF during daytime

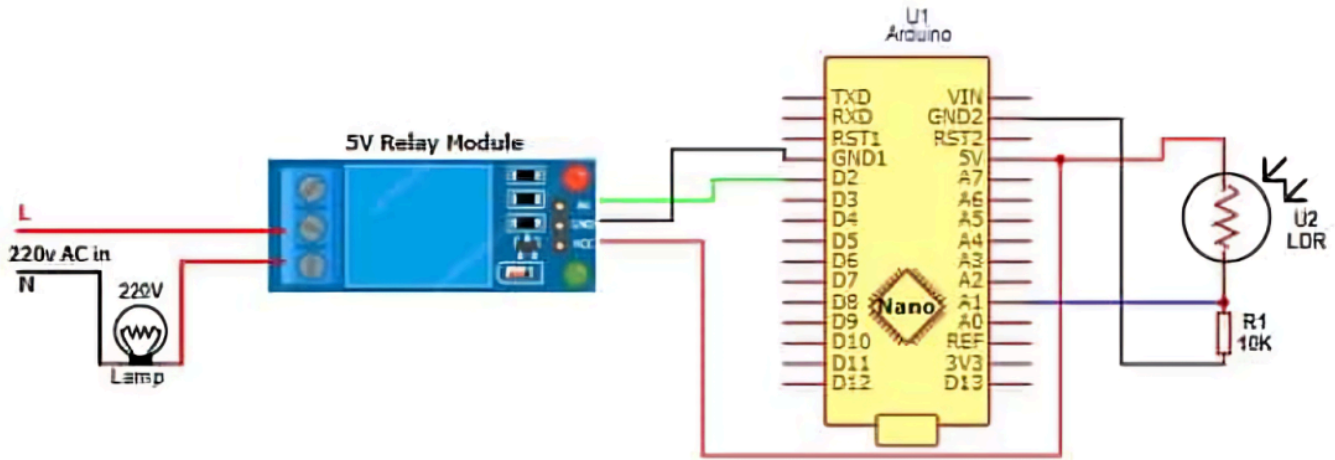
Example:

- Morning time → Sunlight is high → Street light OFF
- Night time → No sunlight → Street light ON
- If someone walks or a vehicle passes → Light becomes brighter (if PIR sensor is used)

6. Power Supply:

- Provides electrical power to all components
- Ensures proper working of the system

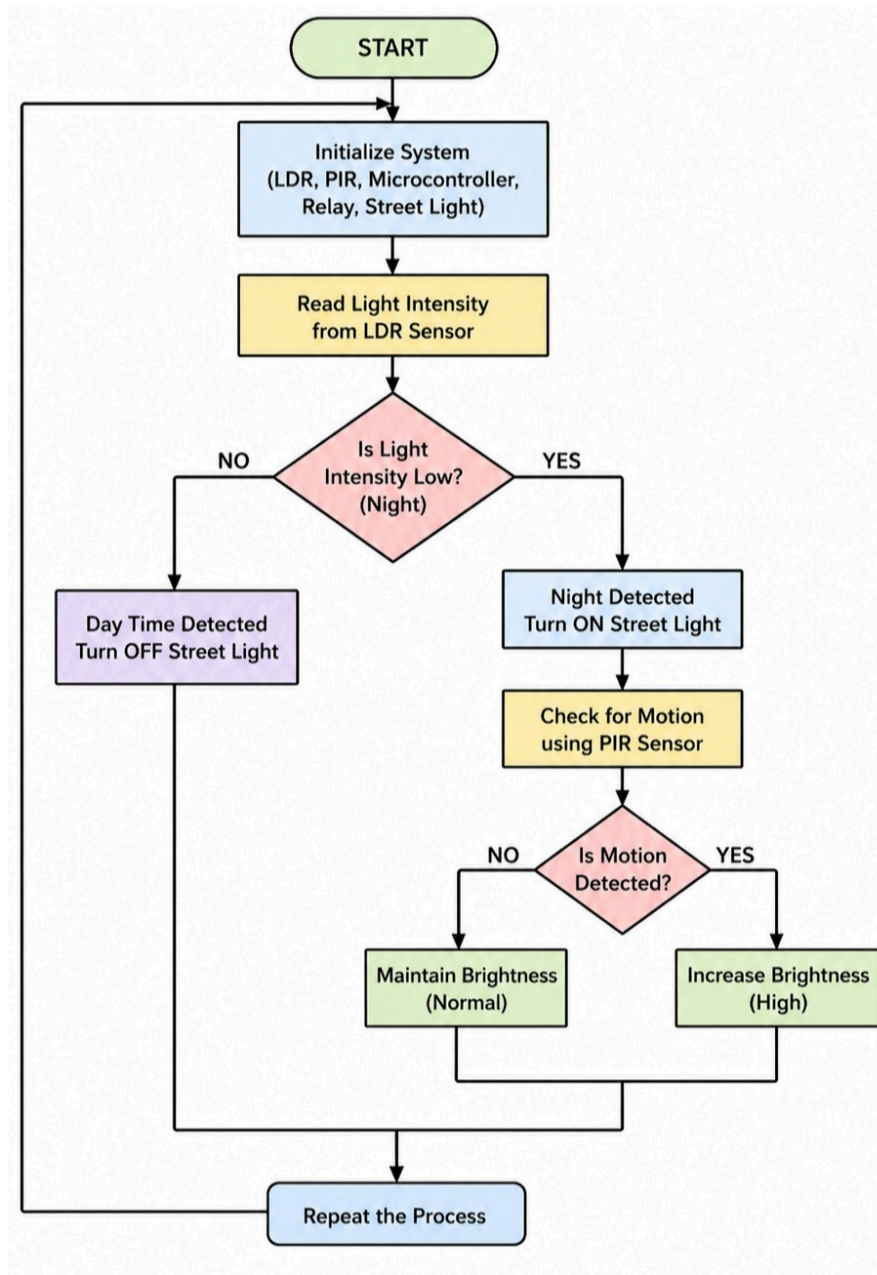
Circuit diagram:



Working Principle :

- The LDR sensor detects the surrounding light intensity
- During daytime, high light is detected by LDR
- The microcontroller receives this signal
- It turns OFF the street light through relay
- During night, low light is detected by LDR
- The microcontroller processes the signal and turns ON the light
- The system works automatically without human intervention

FlowChart :



Program :

Arduino Code :

```

int LDR = A0;
int relay = 8;
int value = 0;
  
```

```

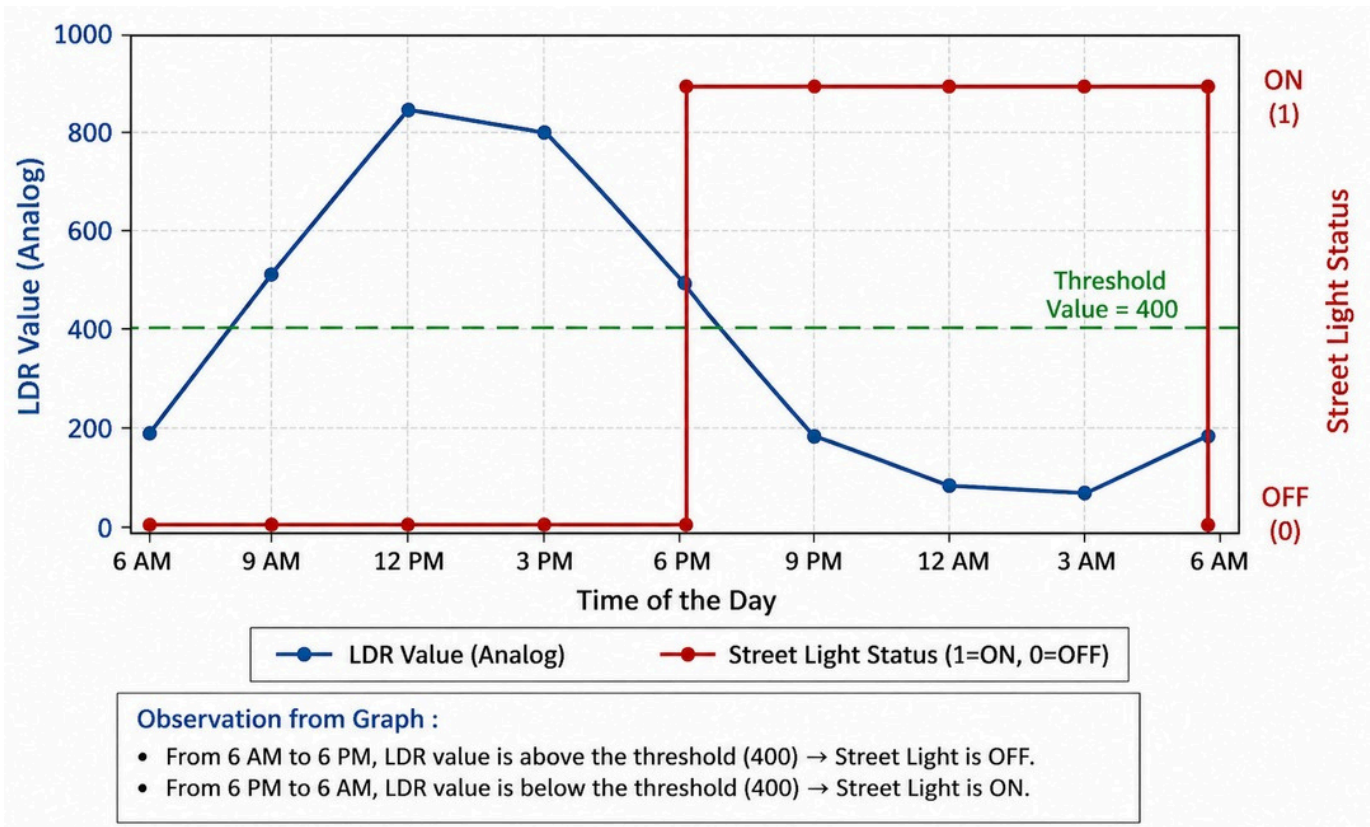
void setup()
{
  pinMode(relay, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  value = analogRead(LDR); // Read LDR value

  if (value < 500)          // Dark condition (night)
  {
    digitalWrite(relay, HIGH); // Turn ON light
  }
  else                      // Bright condition (day)
  {
    digitalWrite(relay, LOW); // Turn OFF light
  }

  delay(500);
}

```

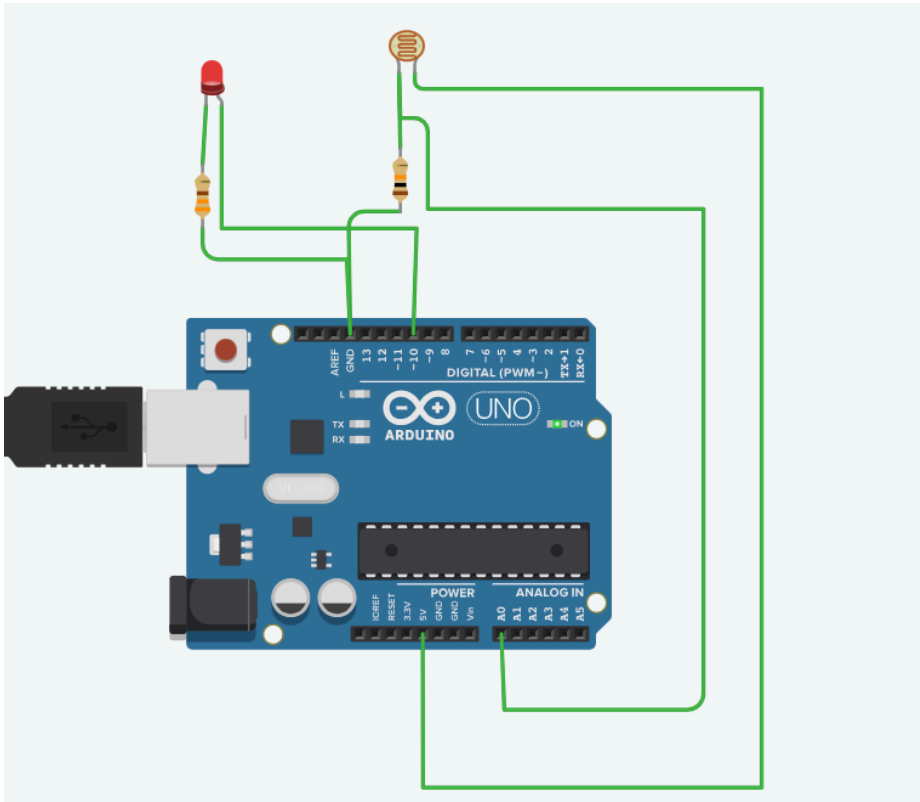
Graph :

Tabular Column:

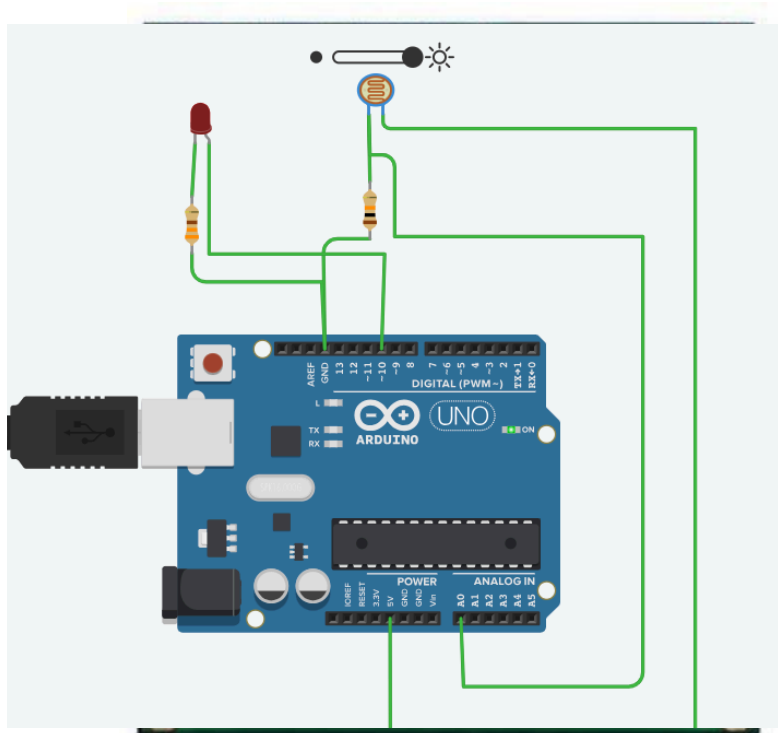
Time	LDR Value	Street Light Status
6 AM	200	OFF
9 AM	500	OFF
12 PM	850	OFF
3 PM	800	OFF
6 PM	450	OFF
9 PM	200	ON
12 AM	100	ON
3 AM	80	ON
6 AM	200	OFF

Output :

a) Night (ON)



d) 75% Level :



Advantages :

- Saves electrical energy
- Reduces manual effort
- Automatic operation
- Low maintenance cost
- Increases efficiency of system
- Environment friendly
- Reliable and easy to use

Applications :

- Street lighting systems
- Highways and roads
- Parking areas
- Garden and campus lighting
- Public places and walkways

Results & Future Enhancement:

◆ **Results :**

- The smart street light system works automatically based on light intensity
- The LDR sensor correctly detects day and night conditions
- The microcontroller controls the street light ON/OFF accurately
- The system reduces energy consumption and electricity cost
- It minimizes manual effort and improves efficiency

◆ Comparison with Other Methods

Algorithm	Accuracy	Complexity
ManualMonitoring	Low	Low
Timer Based	Medium	Medium
Smart System	High	Medium

◆ Future Enhancements :

- Add IoT support for remote monitoring and control
- Use solar panels for energy saving
- Integrate GSM/Wi-Fi module for alerts
- Improve system using advanced sensors (ultrasonic / smart sensors)
- Implement smart city integration

Git Hub Link of the project along with the report	https://github.com/sharavana864/smart-street-lights
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References :

- **Arduino Official Documentation:**
<https://www.arduino.cc/en/Guide>
- **Microcontroller and Embedded Systems:**
<https://www.oreilly.com/library/view/microcontroller-embedded-systems/9789332555408/>
- **LDR Sensor Documentation:**
<https://components101.com/sensors/LDR-sensor-module>
- **Electronics Clinic– street light Project:**
<https://www.electronicclinic.com/smart-street-light-project-using-arduino/>
- **IEEE Research Papers:**
[https://ieeexplore.ieee.org/Smart lighting](https://ieeexplore.ieee.org/Smart%20lighting)